Spyder

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...ers\wamul\OneDrive\Desktop\Computer-Logic-and-Programming

C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled7.py

untitled7.py* | untitled8.py* | untitled9.py* | untitled10.py* | untitled11.py* | untitled12.py

- untitled7.py
  - compute_total
  - main

```python
# -*- coding: utf-8 -*-
"""
Created on Thu Mar 26 17:48:09 2026

@author: wamul
# Problem 1 - Functions Pass By Value
# Quantity and price calculator with 10% discount if total > $10,000


"""

def compute_total(quantity, price):
    # Calculate the initial extended price
    extended_price = quantity * price

    # Check if the total qualifies for a 10% discount
    if extended_price > 10000.00:
        extended_price = extended_price * 0.90

    return extended_price

def main():
    total_extended_price = 0

    while True:
        # Get user input
        qty_input = input("Enter quantity (or '0' to stop): ")
        if qty_input == '0':
            break

        price_input = input("Enter unit price: ")

        # Convert inputs to numbers
        quantity = float(qty_input)
        price = float(price_input)

        # Call the function (Pass by Value)
        item_total = compute_total(quantity, price)

        # Add to the running grand total
        total_extended_price += item_total

        # Display results for this item
        print(f"Qty: {quantity} | Price: ${price:.2f} | Total: ${item_total:.2f}\n")

    # Display final grand total
    print(f"Grand Total for all items: ${total_extended_price:.2f}")

if __name__ == "__main__":
    main()
```

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CIS 106 Problem Set 8 - Data File and For Loops.docx	3/10/2026 10:30
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Console 1/A

```
In [42]: %runfile C:/Users/wamul/OneDrive/Desktop/Computer-Logic-
and-Programming/untitled7.py --wdir
Enter quantity (or '0' to stop): 200
Enter unit price: 25
Qty: 200.0 | Price: $25.00 | Total: $5000.00

Enter quantity (or '0' to stop): 400
Enter unit price: 25
Qty: 400.0 | Price: $25.00 | Total: $10000.00

Enter quantity (or '0' to stop): 500
Enter unit price: 25
Qty: 500.0 | Price: $25.00 | Total: $11250.00

Enter quantity (or '0' to stop): 0
Grand Total for all items: $26250.00

In [43]:
```

IPython Console | History

Inline | Conda: spyder-runtime (Python 3.12.11) | LSP: Python | Line 11, Col 2 | UTF-8 | CRLF | RW | Mem 78%

Spyder

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...ers\wamul\OneDrive\Desktop\Computer-Logic-and-Programming

C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled8.py

untitled7.py* | untitled8.py* | untitled9.py* | untitled10.py* | untitled11.py* | untitled12.py

- untitled8.py
  - compute_batting_average
  - main

```python
@author: wamul
# Problem 2 - Functions Pass By Value
# Baseball batting average calculator
"""

def compute_batting_average(hits, at_bats):
    """Compute batting average: hits divided by at-bats."""
    if at_bats == 0:
        return 0.0
    return hits / at_bats


def main():
    player_count = 0

    print("=== Baseball Batting Average Calculator ===")
    print("Enter 'done' for last name to stop.\n")

    while True:
        last_name = input("Enter player's last name ('done' to stop): ").strip()
        if last_name.lower() == "done":
            break

        try:
            hits = int(input(f"Enter number of hits for {last_name}: "))
            at_bats = int(input(f"Enter number of at-bats for {last_name}: "))
        except ValueError:
            print("Invalid input. Please enter whole numbers for hits and at-bats.")
            continue

        avg = compute_batting_average(hits, at_bats)
        player_count += 1

        print(f"  Player: {last_name}  |  Batting Average: {avg:.3f}\n")

    print("\n" + "=" * 40)
    print(f"Total number of players entered: {player_count}")


if __name__ == "__main__":
    main()
```

Name	Date Modified
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.gitattributes	1/29/2026 12:46
CIS 106 Module 4 Assignments Set Problems.docx	2/13/2026 11:15
CIS 106 Module 12 Array Assignment.docx	3/30/2026 11:39
CIS 106 Module 13 Lists_Dictionary Assignment.docx	3/30/2026 11:55
CIS 106 Module 14 Object Oriented Programming.docx	3/30/2026 11:54
CIS 106 Problem Set 7 - while loops.docx	3/4/2026 9:28 A
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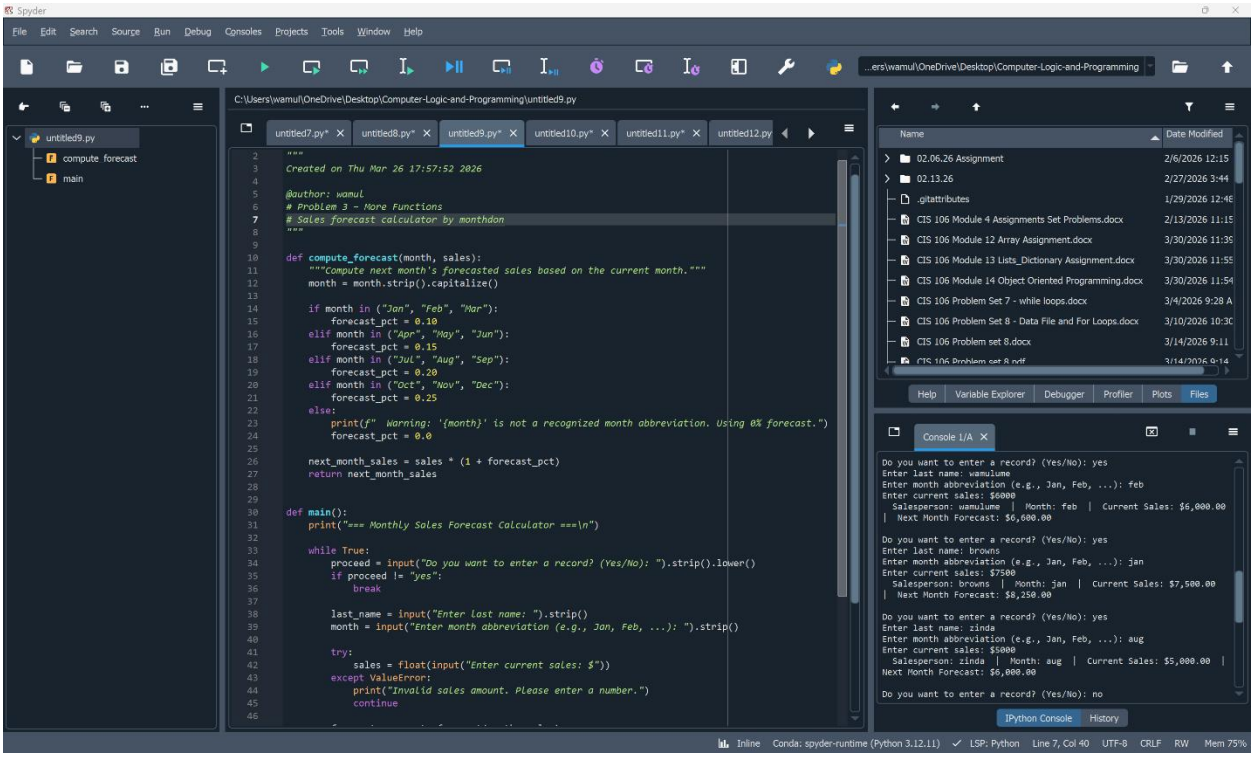
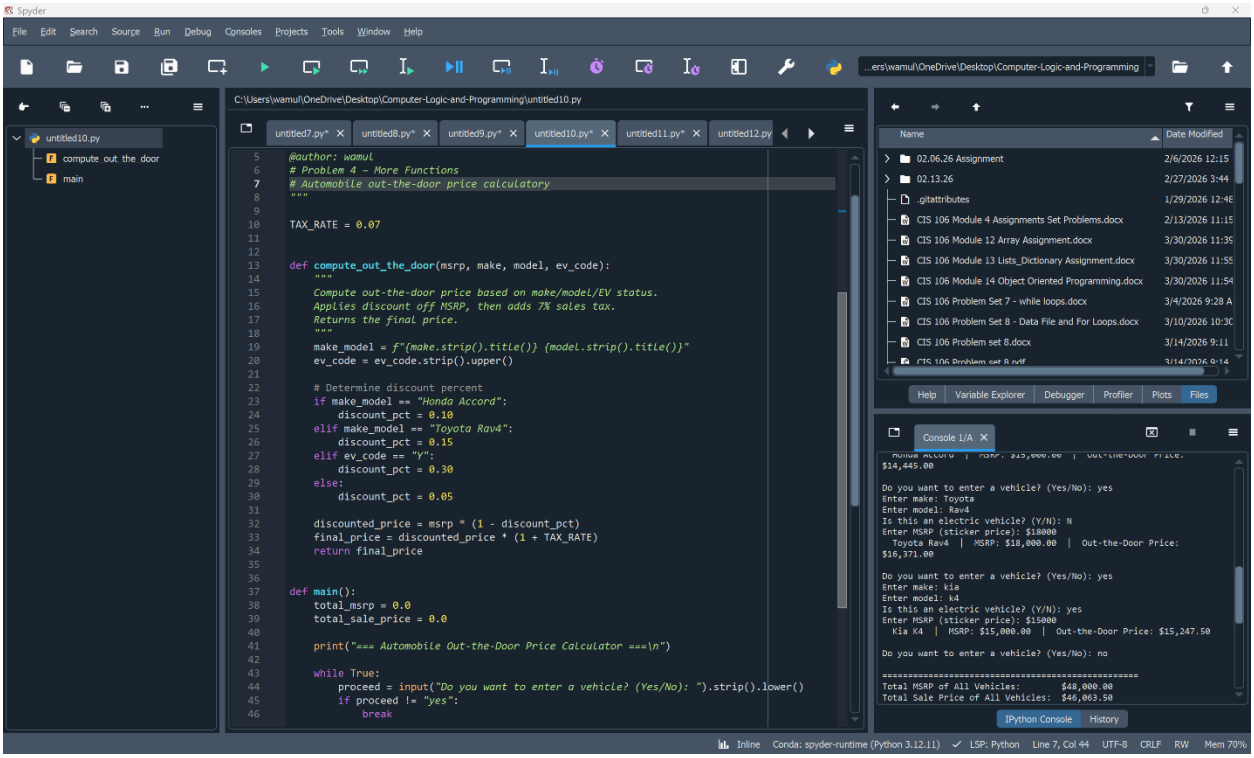
Console 1/A

```
In [43]: %runfile C:/Users/wamul/OneDrive/Desktop/Computer-Logic-
and-Programming/untitled8.py --wdir
=== Baseball Batting Average Calculator ===
Enter 'done' for last name to stop.

Enter player's last name ('done' to stop): wamulume
Enter number of hits for wamulume: 45
Enter number of at-bats for wamulume: 28
  Player: wamulume  |  Batting Average: 1.607

Enter player's last name ('done' to stop): malz
Enter number of hits for malz: 34
Enter number of at-bats for malz: 20
  Player: malz  |  Batting Average: 1.700

Enter player's last name ('done' to stop): done

========================================
Total number of players entered: 2

In [44]:
```

IPython Console | History

Inline | Conda: spyder-runtime (Python 3.12.11) | LSP: Python | Line 15, Col 2 | UTF-8 | CRLF | RW | Mem 75%


Spyder
File Edit Search Source Run Debug Consoles Projects Tools Window Help
C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled9.py
untitled7.py* untitled8.py* untitled9.py* untitled10.py* untitled11.py* untitled12.py
untitled9.py
compute_forecast
main
"""
Created on Thu Mar 26 17:57:52 2026
@author: wamul
Problem 3 - More Functions
Sales forecast calculator by monthdon
"""
def compute_forecast(month, sales):
"""Compute next month's forecasted sales based on the current month."""
month = month.strip().capitalize()
if month in ("Jan", "Feb", "Mar"):
forecast_pct = 0.10
elif month in ("Apr", "May", "Jun"):
forecast_pct = 0.15
elif month in ("Jul", "Aug", "Sep"):
forecast_pct = 0.20
elif month in ("Oct", "Nov", "Dec"):
forecast_pct = 0.25
else:
print(f" Warning: '{month}' is not a recognized month abbreviation. Using 0% forecast.")
forecast_pct = 0.0
next_month_sales = sales * (1 + forecast_pct)
return next_month_sales
def main():
print("=== Monthly Sales Forecast Calculator ===\n")
while True:
proceed = input("Do you want to enter a record? (Yes/No): ").strip().lower()
if proceed != "yes":
break
last_name = input("Enter last name: ").strip()
month = input("Enter month abbreviation (e.g., Jan, Feb, ...): ").strip()
try:
sales = float(input("Enter current sales: $"))
except ValueError:
print("Invalid sales amount. Please enter a number.")
continue
Console 1/A
Do you want to enter a record? (Yes/No): yes
Enter last name: wamulume
Enter month abbreviation (e.g., Jan, Feb, ...): feb
Enter current sales: $6000
Salesperson: wamulume | Month: feb | Current Sales: $6,000.00
| Next Month Forecast: $6,600.00
Do you want to enter a record? (Yes/No): yes
Enter last name: browns
Enter month abbreviation (e.g., Jan, Feb, ...): jan
Enter current sales: $7500
Salesperson: browns | Month: jan | Current Sales: $7,500.00
| Next Month Forecast: $8,250.00
Do you want to enter a record? (Yes/No): yes
Enter last name: zinda
Enter month abbreviation (e.g., Jan, Feb, ...): aug
Enter current sales: $5000
Salesperson: zinda | Month: aug | Current Sales: $5,000.00 |
Next Month Forecast: $6,000.00
Do you want to enter a record? (Yes/No): no
IPython Console History
Inline Conda: spyder-runtime (Python 3.12.11) LSP: Python Line 7, Col 40 UTF-8 CRLF RW Mem 75%


Spyder
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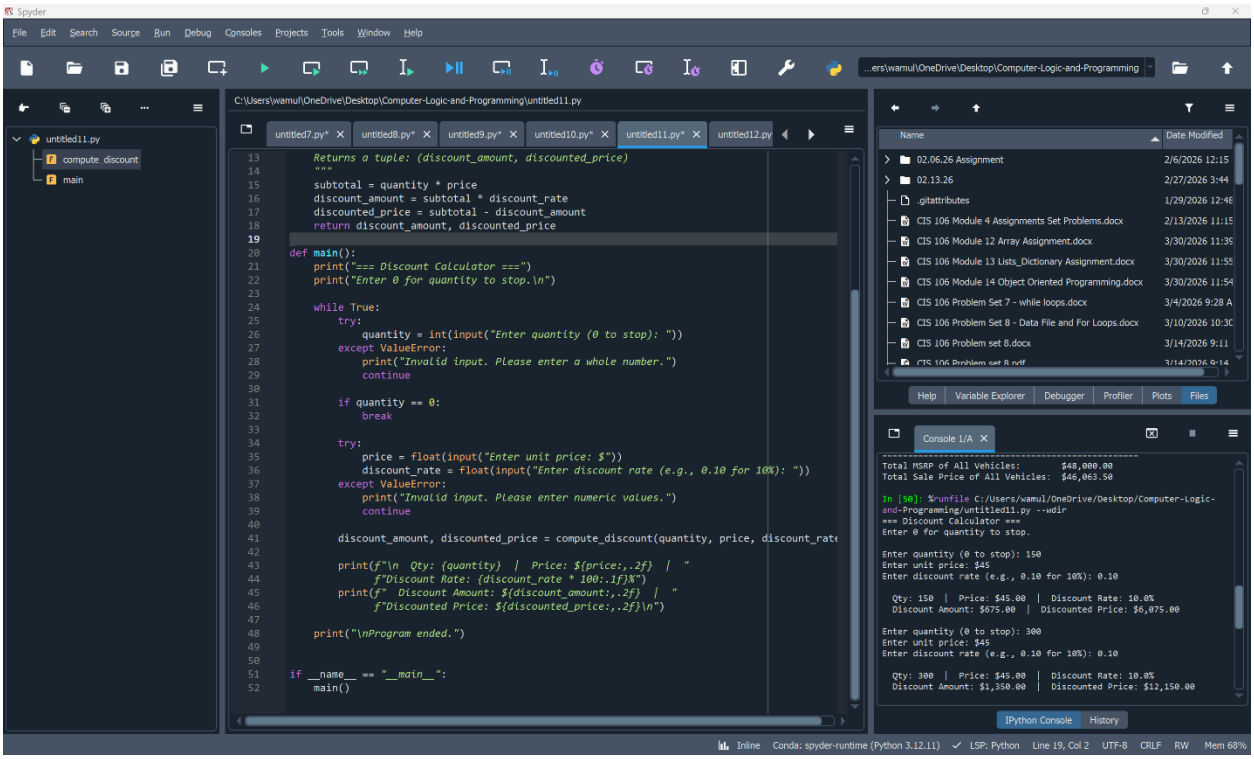
C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled10.py
untitled7.py* untitled8.py* untitled9.py* untitled10.py* untitled11.py* untitled12.py
untitled10.py
compute_out_the_door
main
@author: wamul
Problem 4 - More Functions
Automobile out-the-door price calculatory
"""
TAX_RATE = 0.07
def compute_out_the_door(msrp, make, model, ev_code):
"""
Compute out-the-door price based on make/model/EV status.
Applies discount off MSRP, then adds 7% sales tax.
Returns the final price.
"""
make_model = f"{make.strip().title()} {model.strip().title()}"
ev_code = ev_code.strip().upper()
# Determine discount percent
if make_model == "Honda Accord":
discount_pct = 0.10
elif make_model == "Toyota Rav4":
discount_pct = 0.15
elif ev_code == "Y":
discount_pct = 0.30
else:
discount_pct = 0.05
discounted_price = msrp * (1 - discount_pct)
final_price = discounted_price * (1 + TAX_RATE)
return final_price
def main():
total_msrp = 0.0
total_sale_price = 0.0
print("=== Automobile Out-the-Door Price Calculator ===\n")
while True:
proceed = input("Do you want to enter a vehicle? (Yes/No): ").strip().lower()
if proceed != "yes":
break
Console 1/A
$14,445.00
Do you want to enter a vehicle? (Yes/No): yes
Enter make: Toyota
Enter model: Rav4
Is this an electric vehicle? (Y/N): N
Enter MSRP (sticker price): $18000
Toyota Rav4 | MSRP: $18,000.00 | Out-the-Door Price:
$16,371.00
Do you want to enter a vehicle? (Yes/No): yes
Enter make: kia
Enter model: k4
Is this an electric vehicle? (Y/N): yes
Enter MSRP (sticker price): $15000
Kia K4 | MSRP: $15,000.00 | Out-the-Door Price: $15,247.50
Do you want to enter a vehicle? (Yes/No): no
==========
Total MSRP of All Vehicles: $48,000.00
Total Sale Price of All Vehicles: $46,063.50
IPython Console History
Inline Conda: spyder-runtime (Python 3.12.11) LSP: Python Line 7, Col 44 UTF-8 CRLF RW Mem 70%


Spyder
File Edit Search Source Run Debug Consoles Projects Tools Window Help
C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled11.py
untitled11.py
compute_discount
main
untitled7.py* untitled8.py* untitled9.py* untitled10.py* untitled11.py* untitled12.py
Returns a tuple: (discount_amount, discounted_price)
"""
subtotal = quantity * price
discount_amount = subtotal * discount_rate
discounted_price = subtotal - discount_amount
return discount_amount, discounted_price
def main():
print("=== Discount Calculator ===")
print("Enter 0 for quantity to stop.\n")
while True:
try:
quantity = int(input("Enter quantity (0 to stop): "))
except ValueError:
print("Invalid input. Please enter a whole number.")
continue
if quantity == 0:
break
try:
price = float(input("Enter unit price: $"))
discount_rate = float(input("Enter discount rate (e.g., 0.10 for 10%): "))
except ValueError:
print("Invalid input. Please enter numeric values.")
continue
discount_amount, discounted_price = compute_discount(quantity, price, discount_rate
print(f"\n Qty: {quantity} | Price: ${price:,.2f} | "
f"Discount Rate: {discount_rate * 100:.1f}%")
print(f" Discount Amount: ${discount_amount:,.2f} | "
f"Discounted Price: ${discounted_price:,.2f}\n")
print("\nProgram ended.")
if __name__ == "__main__":
main()
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CIS 106 Problem Set 7 - while loops.docx
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CIS 106 Problem set 8.docx
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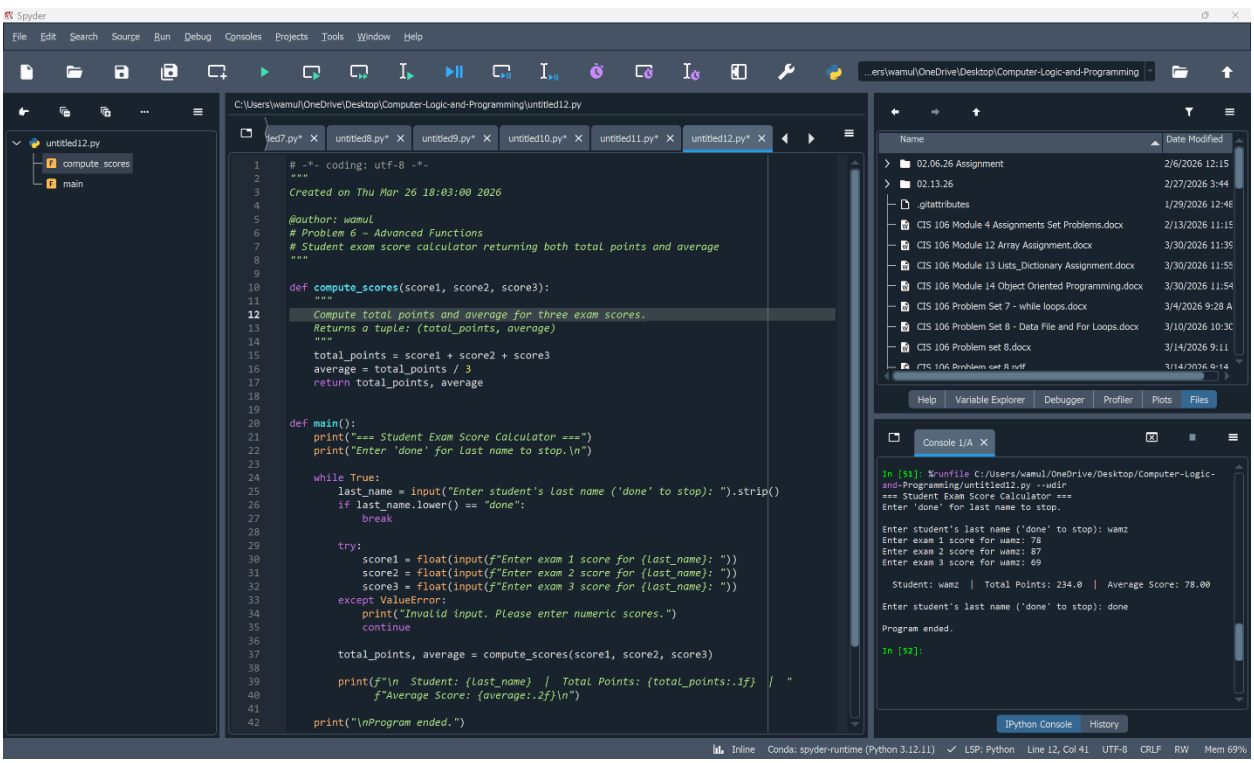
Console 1/A
Total MSRP of All Vehicles: $48,000.00
Total Sale Price of All Vehicles: $46,063.50
In [50]: %runfile C:/Users/wamul/OneDrive/Desktop/Computer-Logic-and-Programming/untitled11.py --wdir
=== Discount Calculator ===
Enter 0 for quantity to stop.
Enter quantity (0 to stop): 150
Enter unit price: $45
Enter discount rate (e.g., 0.10 for 10%): 0.10
Qty: 150 | Price: $45.00 | Discount Rate: 10.0%
Discount Amount: $675.00 | Discounted Price: $6,075.00
Enter quantity (0 to stop): 300
Enter unit price: $45
Enter discount rate (e.g., 0.10 for 10%): 0.10
Qty: 300 | Price: $45.00 | Discount Rate: 10.0%
Discount Amount: $1,350.00 | Discounted Price: $12,150.00
IPython Console History
Inline Conda: spyder-runtime (Python 3.12.11) LSP: Python Line 19, Col 2 UTF-8 CRLF RW Mem 68%


Spyder
File Edit Search Source Run Debug Consoles Projects Tools Window Help
C:\Users\wamul\OneDrive\Desktop\Computer-Logic-and-Programming\untitled12.py
untitled12.py
compute_scores
main
untitled7.py* untitled8.py* untitled9.py* untitled10.py* untitled11.py* untitled12.py*
-*- coding: utf-8 -*-
"""
Created on Thu Mar 26 18:03:00 2026
@author: wamul
Problem 6 – Advanced Functions
Student exam score calculator returning both total points and average
"""
def compute_scores(score1, score2, score3):
"""
Compute total points and average for three exam scores.
Returns a tuple: (total_points, average)
"""
total_points = score1 + score2 + score3
average = total_points / 3
return total_points, average
def main():
print("=== Student Exam Score Calculator ===")
print("Enter 'done' for last name to stop.\n")
while True:
last_name = input("Enter student's last name ('done' to stop): ").strip()
if last_name.lower() == "done":
break
try:
score1 = float(input(f"Enter exam 1 score for {last_name}: "))
score2 = float(input(f"Enter exam 2 score for {last_name}: "))
score3 = float(input(f"Enter exam 3 score for {last_name}: "))
except ValueError:
print("Invalid input. Please enter numeric scores.")
continue
total_points, average = compute_scores(score1, score2, score3)
print(f"\n Student: {last_name} | Total Points: {total_points:.1f} | "
f"Average Score: {average:.2f}\n")
print("\nProgram ended.")
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CIS 106 Module 13 Lists_Dictionary Assignment.docx
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CIS 106 Problem Set 7 - while loops.docx
CIS 106 Problem Set 8 - Data File and For Loops.docx
CIS 106 Problem set 8.docx
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Console 1/A
In [51]: %runfile C:/Users/wamul/OneDrive/Desktop/Computer-Logic-and-Programming/untitled12.py --wdir
=== Student Exam Score Calculator ===
Enter 'done' for last name to stop.
Enter student's last name ('done' to stop): wamz
Enter exam 1 score for wamz: 78
Enter exam 2 score for wamz: 87
Enter exam 3 score for wamz: 69
Student: wamz | Total Points: 234.0 | Average Score: 78.00
Enter student's last name ('done' to stop): done
Program ended.
In [52]:
IPython Console History
Inline Conda: spyder-runtime (Python 3.12.11) LSP: Python Line 12, Col 41 UTF-8 CRLF RW Mem 69%